```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import make_regression
from sklearn.model_selection import train_test_split
from sklearn.linear_model import Ridge
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import mean_squared_error, r2_score

# 1. Generate a synthetic dataset
X, y = make_regression(n_samples=100, n_features=1, noise=20, random_state=42)

# 2. Split the data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# 3. Scale the features (Ridge is sensitive to scale)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# 4. Initialize and fit the Ridge Regression model
# alpha is the regularization strength
ridge_reg = Ridge(alpha=1.0)
ridge_reg.fit(X_train_scaled, y_train)

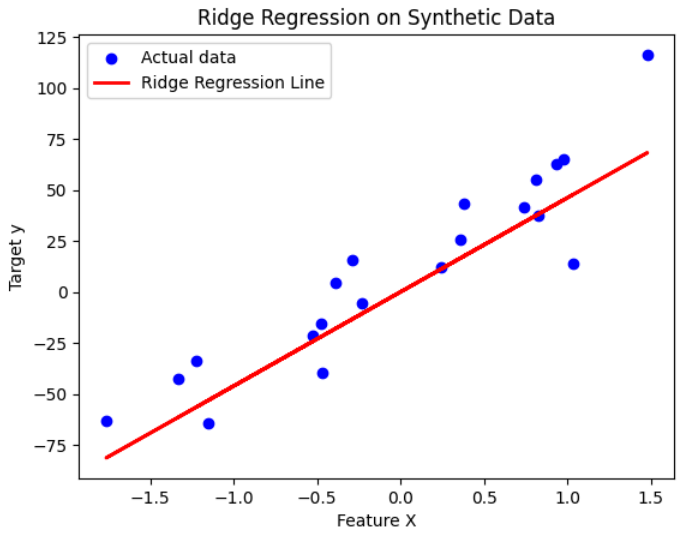
# 5. Make predictions
y_pred = ridge_reg.predict(X_test_scaled)

# 6. Evaluate the model
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print(f"Mean Squared Error: {mse:.2f}")
print(f"R^2 Score: {r2:.2f}")
print(f"Coefficients: {ridge_reg.coef_}")
print(f"Intercept: {ridge_reg.intercept_:.2f}")
```

```
Mean Squared Error: 419.20
R^2 Score: 0.80
Coefficients: [41.75259907]
Intercept: -5.82
```

```
# 7. Visualize the results
plt.scatter(X_test, y_test, color='blue', label='Actual data')
plt.plot(X_test, y_pred, color='red', linewidth=2, label='Ridge Regression Line')
plt.title('Ridge Regression on Synthetic Data')
plt.xlabel('Feature X')
plt.ylabel('Target y')
plt.legend()
plt.show()
```



Exporting to PDF

If the browser print-to-PDF failed to include your graphs, run the following cells to generate a PDF using LaTeX conversion.

```
# 1. Update package lists and install dependencies
!apt-get update
!apt-get install -y texlive texlive-xetex texlive-latex-extra pandoc
!pip install pypandoc
```

```
Hit:1 http://archive.ubuntu.com/ubuntu jammy InRelease
Get:2 https://cli.github.com/packages stable InRelease [3,917 B]
Get:3 http://archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
```

```
Get:4 https://cloud.r-project.org/bin/linux/ubuntu jammy-cran40/ InRelease [3,632 B]
Get:5 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Get:6 https://r2u.stat.illinois.edu/ubuntu jammy InRelease [6,555 B]
Get:7 http://archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:8 https://cli.github.com/packages stable/main amd64 Packages [356 B]
Get:9 https://cloud.r-project.org/bin/linux/ubuntu jammy-cran40/ Packages [107 kB]
Get:10 https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu jammy InRelease [18.1 kB]
Get:11 https://r2u.stat.illinois.edu/ubuntu jammy/main amd64 Packages [3,164 kB]
Hit:12 https://ppa.launchpadcontent.net/ubuntugis/ppa/ubuntu jammy InRelease
Get:13 http://archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 Packages [7,836 kB]
Get:14 https://r2u.stat.illinois.edu/ubuntu jammy/main all Packages [10.7 MB]
Get:15 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [4,524 kB]
Get:16 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 Packages [1,616 kB]
Get:17 http://archive.ubuntu.com/ubuntu jammy-updates/multiverse amd64 Packages [92.7 kB]
Get:18 http://archive.ubuntu.com/ubuntu jammy-backports/main amd64 Packages [82.8 kB]
Get:19 http://archive.ubuntu.com/ubuntu jammy-backports/universe amd64 Packages [35.6 kB]
Get:20 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [4,190 kB]
Get:21 https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu jammy/main amd64 Packages [44.3 kB]
Get:22 http://security.ubuntu.com/ubuntu jammy-security/multiverse amd64 Packages [84.1 kB]
Get:23 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 Packages [1,314 kB]
Get:24 http://security.ubuntu.com/ubuntu jammy-security/restricted amd64 Packages [7,545 kB]
Fetched 41.8 MB in 4s (11.3 MB/s)
Reading package lists... Done
W: Skipping acquire of configured file 'main/source/Sources' as repository 'https://r2u.stat.illinois.edu/ubuntu jammy InRelease' does not seem to provide it (
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
pandoc is already the newest version (2.9.2.1-3ubuntu2).
pandoc set to manually installed.
The following additional packages will be installed:
  dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
  fonts-texgyre fonts-urw-base35 libapache-pom-java libcommons-logging-java
  libcommons-parent-java libfontbox-java libgs9 libgs9-common libidn12
  libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java libptexenc1 libruby3.0
  libsynctex2 libteckit0 libtexlua53 libtexluajit2 libwoff1 libzzip-0-13
  lmodern poppler-data preview-latex-style rake ruby ruby-net-telnet
  ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0 rubygems-integration t1utils
  teckit tex-common tex-gyre texlive-base texlive-binaries
  texlive-fonts-recommended texlive-latex-base texlive-latex-recommended
  texlive-pictures texlive-plain-generic tipa xfonts-encodings xfonts-utils
Suggested packages:
  fonts-noto fonts-freefont-otf | fonts-freefont-ttf libavalon-framework-java
  libcommons-logging-java-doc libexcalibur-logkit-java liblog4j1.2-java
  poppler-utils ghostscript fonts-japanese-mincho | fonts-ipafont-mincho
  fonts-japanese-gothic | fonts-ipafont-gothic fonts-arphic-ukai
  fonts-arphic-uming fonts-nanum ri ruby-dev bundler debhelper gv
  | postscript-viewer perl-tk xpdf | pdf-viewer xzdec
  texlive-fonts-recommended-doc texlive-latex-base-doc python3-pygments
  icc-profiles libfile-which-perl libspreadsheet-parseexcel-perl
  texlive-latex-extra-doc texlive-latex-recommended-doc texlive-luatex
  texlive-pstricks dot2tex prerex texlive-pictures-doc vprerex
  default-jre-headless tipa-doc
The following NEW packages will be installed:
  dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
```

```
import nbformat as nbf
import os

# Create a new notebook object to ensure content is captured
nb = nbf.v4.new_notebook()

# Define the cells we want to include in the report
cells = [
    nbf.v4.new_markdown_cell('# Ridge Regression Analysis Report'),
    nbf.v4.new_code_cell(open('/content/0663c6bf.py', 'r').read() if os.path.exists('/content/0663c6bf.py') else "# Ridge Regression Model Training\n(Code from p
    nbf.v4.new_code_cell("# Visualize the results\n" + open('/content/04e71a31.py', 'r').read() if os.path.exists('/content/04e71a31.py') else "# Visualization C
]

# Since I cannot read the internal cell contents directly after a failure,
# I will manually reconstruct the final report script for you to ensure it works.
report_code = """
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import make_regression
from sklearn.model_selection import train_test_split
from sklearn.linear_model import Ridge
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import mean_squared_error, r2_score

# Generate and Model
X, y = make_regression(n_samples=100, n_features=1, noise=20, random_state=42)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
ridge_reg = Ridge(alpha=1.0)
ridge_reg.fit(X_train_scaled, y_train)
y_pred = ridge_reg.predict(X_test_scaled)

print(f"MSE: {mean_squared_error(y_test, y_pred):.2f}")
print(f"R2: {r2_score(y_test, y_pred):.2f}")

plt.scatter(X_test, y_test, color='blue')
plt.plot(X_test, y_pred, color='red')
plt.title('Ridge Regression Results')
plt.show()
```

```
"""

with open('report.ipynb', 'w') as f:
    nb['cells'] = [nbf.v4.new_code_cell(report_code)]
    nbf.write(nb, f)

print("Converting to PDF...")
!jupyter nbconvert --to pdf report.ipynb --output ridge_regression_report.pdf
print("Done! Download 'ridge_regression_report.pdf' from the file pane.")
```

```
Converting to PDF...
[NbConvertApp] Converting notebook report.ipynb to pdf
[NbConvertApp] Writing 23995 bytes to notebook.tex
[NbConvertApp] Building PDF
[NbConvertApp] Running xelatex 3 times: ['xelatex', 'notebook.tex', '-quiet']
[NbConvertApp] Running bibtex 1 time: ['bibtex', 'notebook']
[NbConvertApp] WARNING | bibtex had problems, most likely because there were no citations
[NbConvertApp] PDF successfully created
[NbConvertApp] Writing 20480 bytes to ridge_regression_report.pdf
Done! Download 'ridge_regression_report.pdf' from the file pane.
```